SATGURU STUDY CENTRE
Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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CLASS : 10

Register Number: 1 0 3 0 2

SECOND REVISION EXAMINATION - 2025

Time Allowed : 3.00 Hours]

MATHEMATICS

[Max. Marks : 100

PART – I

I. Answer all of the following: **14x1=14**

1. A= {a,b,p} , B= {2,3} , C = {p,q,r,s} then n[(A∪C)xB] is
   a) 8 b) 20 c) 12 d) 16
2. If $f(x) = 2x^2$ and $g(x) = \frac{1}{3x}$ then fog is
   a) $\frac{3}{2x^2}$ b) $\frac{2}{3x^2}$ c) $\frac{2}{9x^2}$ d) $\frac{1}{6x^2}$
3. $7^{4k} \equiv$ --------------- (mod 100)
   a) 1 b) 2 c) 3 d) 4
4. The value of $(1^3 + 2^3 + 3^3 + \dots + 15^3) - (1+2+3+ \dots +15)$ is
   a) 14400 b) 14200 c) 14280 d) 14520
5. The solution of $(2x-1)^2 = 9$ is equal to
   a) -1 b) 2 c) -1, 2 d) None of these
6. Transpose of a column matrix is
   a) unit matrix b) diagonal matrix c) column matrix d) row matrix
7. A tangent is perpendicular to the radius at the
   a) centre b) point of contact c) infinity d) chord
8. The point of intersection of $3x-y=4$ and $x+y=8$ is
   a) (5, 3) b) (2, 4) c) (3, 5) d) (4, 4)
9. When Proving that a quadrilateral is a trapezium it is necessary to show
   a) Two sides are Parallel b) Two Parallel and two Parallel sides
   c) Opposite sides are Parallel d) All sides are of equal length
10. a cotθ + b cosecθ =P and b cotθ + a cosecθ = q then p^2-q^2 is equal to
    a) $a^2 - b^2$ b) $b^2 - a^2$ c) $a^2 + b^2$ d) b -a
11. The total surface area of a cylinder whose radius is $\frac{1}{3}$ of its height is
    a) $\frac{9\pi h^2}{8}$ sq.units b) $24\pi h^2$ sq.units c) $\frac{8\pi h^2}{9}$ sq.units d) $\frac{56\pi h^2}{9}$ sq.units
12. A Spherical ball of radius r_1, Units is melted to make 8 new identical balls each of radius r_2 units, Then $r_1 : r_2$ is
    a) 2 : 1 b) 1 : 2 c) 4 : 1 d) 1 : 4
13. Variance of first 20 natural numbers is
    a) 32.25 b) 44.25 c) 33.25 d) 30
14. Which of the following is incorrect?
    a) P(A)>1 b) $0 \le P(A) \le 1$ c) $P(\phi)=0$ d) $P(A)+P(\bar{A})=1$

PART - II

II. Answer any 10 questions. Question No. 28 is compulsory. **10x2=20**

15. If BxA = {(-2, 3), (-2, 4), (0, 3), (0, 4), (3, 3), (3, 4)} find A and B.
16. Given $f(x) = 2x-x^2$ find f(1), and $f(x+1)$.
17. If $13824 = 2^a \times 3^b$ then find a and b.
18. Find the 8th term of the G.P. 9,3,1, ...
19. Determine the nature of the roots $2x^2-2x+9=0$.

20. Construct a 3x3 matrix whose elements are given by $a_{ij} = |i - 2j|$.

21. A man goes 18m due east and then 24m due north. Find the distance of his current position from the starting point?

22. Find the equation of a line whose intercepts on the x and y axes are 4, -6.

23. Show that the straight lines $x-2y+3=0$ and $6x+3y+8=0$ are perpendicular.

24. If the circumference of a conical wooden Piece is 484cm then find its volume when its height is 105cm.

25. The slant height of a frustum of a cone is 5cm and the radii of its ends are 4cm and 1cm. Find its curved surface area.

26. Find the standard deriation of first 21 natural number.

27. Two coins are tossed together. What is the Probability of getting different faces on the coins.

28. Prove that $\frac{\text{cosec}\theta}{\cos\theta} - \frac{\cos\theta}{\sin\theta} = \tan\theta$

PART - III

III. Answer the following any 10 questions. Q.No.42 is compulsory. **10x5=50**

29. Let $A = \{x \in W / x<2\}$, $B = \{x \in N / 1<x\leq 4\}$ and $C=\{3, 5\}$ verify that $(A\cup B)\times C = (A\times C)\cup(B\times C)$.

30. If $f(x)=2x+3$, $g(x)=1-2x$, and $h(x)=3x$ Prove that $fo(goh)=(fog)oh$.

31. In an A.P. sum of four consecutive terms is 28 and their sum of their squares is 276. Find the four number.

32. If $P_1^{x_1} \times P_x^{x_2} \times P_3^{x_3} \times P_4^{x_4} = 113400$ where P_1, P_2, P_3, P_4 are Primes in ascending order and x_1, x_2, x_3, x_4 are integers. Find the value of P_1, P_2, P_3, P_4 and x_1, x_2, x_3, x_4

33. Find the GCD of $6x^3-30x^2+60x-48$ and $3x^3-12x^2+21x-18$.

34. The roots of the equation $x^2+6x-4=0$ are α,β. Find the quadratic equation whose roots are α^2 and β^2

35. If $A = \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$ verify that $(AB)^T = B^TA^T$.

36. Find the area of quadrilateral formed by the points (8, 6), (5, 11), (-5, 12) and (-4, 3)

37. Find the equation of the median of ΔABC through A where the vertices are (6, 2), (-5, -1) and (1, 9).

38. From a Point on the ground, the angle of elevation of the bottom and top of a tower fixed at the top of a 30m high building are 45° and 60° respectively. Find the height of the tower. ($\sqrt{3}=1.732$)

39. The internal and external diameter of a hollow hemispherical shell are 6 cm and 10 cm respectively. If it is melted and recast into a solid cylinder of diameter 14 cm, then find the height of the cylinder.

40. The rainfall recorded in various places of five districts in a week are given below.

Rainfall (in mm) :	45	50	55	60	65	70
No.of.Places :	5	13	4	9	5	4

Find its standard deviation.

41. Two dice are rolled together. Find the Probability of getting a doublet or sum of faces as 4.

42. State and Prove ~~Converse~~ of Angle Bisector Theorem.

PART - IV

IV. Answer the following. **2x8=16**

43. a) Construct a triangle simillar to a given triangle PQR with its sides equal to $^7/_4$ of the corresponding sides of the triangle PQR (Scale factor $^7/_4>1$)

(OR)

b) Construct a ΔPQR with the base PQ=4.5cm, $\angle R=35°$ and the median from R to RG is 6cm.

44. a) Draw the Graph of $y=x^2-5x-6$ and hence solve $x^2-5x-14=0$

(OR)

b) Draw the Graph of $xy=24$, $x, y>0$. Using Graph find (i) y when $x=3$ and (ii) x when $y=6$.